Monotonicity of Turán determinants for binary Krawtchouk polynomials

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Abstract. Let $M, Q \geq 0$ be integers, $D = M + Q$, and let $p_s = [w^s]\,(1+w)^M(1-w)^Q$, equivalently the binary Krawtchouk value $K_s(Q; D)$ in its generating-function normalization. We prove

$$p_s^2 - p_{s-1}p_{s+1} \;\geq\; p_{s+1}^2 - p_s p_{s+2} \qquad (\lceil D/2 \rceil \leq s \leq D).$$

Thus the Turán determinants are symmetric and weakly unimodal in the coefficient index, and they are at least 1 at every index. Equivalently, these determinants are the rectangular Schur values $s_{(2^s)}(1^M, (-1)^Q)$, so the theorem gives positivity and unimodality for a family of GL_D-character values at involutions. This is a computer-assisted proof. Recurrence and comparison arguments cover three infinite parameter regions; exact symbolic certificates cover the remaining finite polynomial families; and an exhaustive integer computation discharges the finite remainders required by the symbolic bounds. The certificate-generating programs and exact replay logs accompany the manuscript.

Keywords. Krawtchouk polynomials, Turán determinant, Schur function, character value, unimodality, three-term recurrence, exact computation

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1. Introduction and main theorem

1.1. The theorem

Fix integers $M, Q \geq 0$, set $D = M + Q$, and define the integer sequence

$$p_s \;=\; [w^s]\,(1+w)^M(1-w)^Q, \qquad s = 0, 1, \ldots, D, \tag{1.1}$$

with $p_s = 0$ outside $0 \leq s \leq D$. These are binary Krawtchouk values: with the standard generating function $\sum_k K_k(x; N)\,w^k = (1+w)^{N-x}(1-w)^x$ one has $p_s = K_s(Q; D)$, the degree-s binary (symmetric, $p = \frac{1}{2}$) Krawtchouk polynomial of length D evaluated at the integer point Q. Define the *Turán determinant* and the *Turán step* of this sequence,

$$\begin{aligned} \mathrm{Tur}(s) &= p_s^2 - p_{s-1}p_{s+1}, \\ T_s &= \mathrm{Tur}(s) - \mathrm{Tur}(s+1) \;=\; p_s^2 + p_s p_{s+2} - p_{s+1}^2 - p_{s-1}p_{s+1}. \end{aligned} \tag{1.2}$$

Theorem 1.1 *(Main Theorem).* *For all integers $M, Q \geq 0$ and s with $\lceil D/2 \rceil \leq s \leq D$,*

$$T_s \;\geq\; 0.$$

Equivalently, $s \mapsto \mathrm{Tur}(s)$ is non-increasing on the top half. Since $p_{D-s} = (-1)^Q p_s$, the sequence $\mathrm{Tur}(s)$ is symmetric about $D/2$ and therefore weakly unimodal.

Two immediate corollaries, both obtained by telescoping from the top end ($p_D = (-1)^Q$, $p_{D+1} = 0$, so $\mathrm{Tur}(D) = 1$):

Corollary 1.2. *For all $M, Q \geq 0$:*

1. *$\mathrm{Tur}(s) \geq 1$ for every $0 \leq s \leq D$, with sharp constant 1;*
2. *$\mathrm{Tur}(s) \leq \mathrm{Tur}(\lceil D/2 \rceil)$ for $0 \leq s \leq D$.*

Corollary 1.3 *(adjacent coefficient lattices).* *For $0 \leq s \leq D$, the two adjacent coefficient vectors*

$$v_s \;=\; (p_{s-1}, p_s), \qquad v_{s+1} \;=\; (p_s, p_{s+1})$$

span a full-rank sublattice $L_s \subseteq \mathbb{Z}^2$ of index $[\mathbb{Z}^2 : L_s] = \mathrm{Tur}(s)$. These lattice indices are symmetric and weakly increase from either endpoint toward the center.

Proof. The absolute determinant of the matrix with columns v_s, v_{s+1} is $|p_{s-1}p_{s+1} - p_s^2| = \mathrm{Tur}(s)$ by Corollary 1.2. The asserted ordering is the Main Theorem and reflection. ∎

Classical Turán inequalities concern the sign of $P_n(x)^2 - P_{n-1}(x)P_{n+1}(x)$ for a fixed degree window and an argument x in the orthogonality interval [Sze48]. Theorem 1.1 instead concerns monotonicity in the degree index (equivalently, the generating-function coefficient index) at a fixed integer argument. The integer restriction is essential: the corresponding real relaxation need not be nonnegative. More precisely, if the integer argument Q is replaced by a real $q \in [0, D]$ and the coefficients are defined by the same formal generating series

$$[w^s](1+w)^{D-q}(1-w)^q,$$

then at $(D, q, s) = (4, \frac{8}{7}, 2)$ one obtains $T_2 = -1430/117649 < 0$. Section 8 gives a more detailed comparison with the literature.

1.2. Schur-character interpretation

Let

$$z \;=\; (\underbrace{1, \ldots, 1}_{M}, \underbrace{-1, \ldots, -1}_{Q}),$$

and write $e_r(z)$ and $s_\lambda(z)$ for the elementary and Schur symmetric functions, respectively. As usual, (2^s) denotes the rectangular partition with s rows of length 2.

Proposition 1.4 *(Schur-character interpretation)*. *For every $0 \le s \le D$,*

$$\mathrm{Tur}(s) \;=\; s_{(2^s)}(z). \tag{1.3}$$

When $D \ge 1$, this is the irreducible polynomial-character value

$$\chi_{(2^s)}(\mathrm{diag}(I_M, -I_Q))\,.$$

The rectangle-complement identity gives $s_{(2^s)}(z) = s_{(2^{D-s})}(z)$ independently of the Main Theorem.

Proof. The generating function in (1.1) is $\prod_{i=1}^{D}(1 + z_i w)$, so $p_r = e_r(z)$. Since the conjugate of (2^s) is (s, s), the dual Jacobi–Trudi identity gives

$$s_{(2^s)}(z) \;=\; \det\begin{pmatrix} e_s(z) & e_{s+1}(z) \\ e_{s-1}(z) & e_s(z) \end{pmatrix} \;=\; p_s^2 - p_{s-1}p_{s+1}.$$

Schur polynomials are the irreducible polynomial characters of the general linear group; see [Mac95, Chapter I, Section 3]. Finally, for a partition λ inside the $2 \times D$ rectangle, with rectangle complement λ^c,

$$s_\lambda(z_1^{-1}, \dots, z_D^{-1}) \;=\; (z_1 \cdots z_D)^{-2} s_{\lambda^c}(z_1, \dots, z_D).$$

Here $z_i^{-1} = z_i$, $(z_1 \cdots z_D)^2 = 1$, and the complement of (2^s) is (2^{D-s}), proving the symmetry. ∎

Thus Theorem 1.1 and Corollary 1.2 say that the rectangular character values in (1.3) are positive integers and form a symmetric weakly unimodal sequence as the rectangle height s varies. The character interpretation alone does not imply the sign: the involution has negative eigenvalues, so ordinary Schur positivity does not apply to this specialization. It does, however, identify a canonical representation-theoretic object and suggests seeking a cancellation-free branching or restriction interpretation of the ordered differences T_s.

1.3. Why this minor sequence is natural

For a fixed integer argument Q, the values $(p_s)_{s=0}^{D}$ form a row of the binary Krawtchouk matrix. The quantity

$$\mathrm{Tur}(s) \;=\; \det\begin{pmatrix} p_s & p_{s+1} \\ p_{s-1} & p_s \end{pmatrix}$$

is its adjacent 2×2 Toeplitz minor. The theorem therefore orders all such adjacent minors along half of a Krawtchouk row; reflection orders the other half. In particular, every one of these signed-row minors is a positive integer at least 1, even though the row itself oscillates and may contain zeros. Corollary 1.3 gives an equivalent arithmetic interpretation: the minors are the indices of the lattices generated by successive coefficient vectors. This makes the statement stronger than a pointwise Turán sign inequality and places it naturally beside minor and kernel inequalities used in Hamming-scheme spectral analysis. The exact generating function, standard recurrences, and normalizations for Krawtchouk polynomials may be found in [KLS10, Section 9.11].

The endpoint pairs are unimodular: $\mathrm{Tur}(0) = \mathrm{Tur}(D) = 1$. The Main Theorem therefore gives an exact endpoint minimum and a global ordering for every row, not just nonvanishing of its adjacent minors: the covolume is smallest, and equal to one, at both endpoints and never decreases while moving toward the center. We do not claim that this observation alone improves a coding bound; its role here is to identify the arithmetic content of the ordered-minor statement and the normalization in which the sharp endpoint value is meaningful.

A small row illustrates all features at once. For $(M, Q) = (5, 2)$,

$$(p_s)_{s=0}^{7} = (1, 3, 1, -5, -5, 1, 3, 1), \qquad \big(\mathrm{Tur}(s)\big)_{s=0}^{7} = (1, 8, 16, 30, 30, 16, 8, 1).$$

Thus the coefficients oscillate, while the adjacent minors remain positive, symmetric, and ordered toward the center.

1.4. Normalization

The normalization in (1.1) is part of the statement. If $q_s = c_s p_s$, then

$$q_s^2 - q_{s-1}q_{s+1} = c_s^2 p_s^2 - c_{s-1}c_{s+1}p_{s-1}p_{s+1}.$$

Thus a Turán determinant is merely rescaled by a positive factor for every sequence p only when $c_s^2 = c_{s-1}c_{s+1}$, that is, when (c_s) is geometric. Degree-dependent orthonormal or endpoint normalizations of Krawtchouk polynomials therefore do not preserve Theorem 1.1.

The coefficient normalization is natural here because it is integral at the integer arguments under consideration, it has the exact reflection $p_{D-s} = (-1)^Q p_s$, and it fixes the endpoint $\mathrm{Tur}(D) = p_D^2 = 1$. The last two features turn top-half monotonicity into symmetric weak unimodality and the sharp lower bound of Corollary 1.2. Comparisons with normalized Turán inequalities in Section 8 are therefore comparisons of phenomena, not claims of invariance under a change to the present normalization.

1.5. Guide to the proof

The top half $\{2s \geq D\}$ is divided into the bulk, flow, center, and gap regions. A positive-semidefinite quadratic form treats the bulk, and a Riccati comparison treats the flow region. The remaining gap is handled by fixed-offset certificates, fixed-argument certificates, and two parity-dependent recurrence arguments. Figure 1 and Table 1 summarize the decomposition; the proof is assembled in Section 6.2.

We assume $x = M - Q \geq 0$ without loss of generality. Indeed, T_s is invariant under $M \leftrightarrow Q$: the substitution $w \mapsto -w$ sends p_s to $(-1)^s p_s$, and every product in T_s has even index sum.

1.6. Proof status and computer assistance

Three kinds of argument occur below. First, the bulk, center, flow, and even-minimum regions have proofs in the text; their accompanying programs replay algebraic identities and finite diagnostics but do not replace the written implications. Second, the finite-offset families, the fixed-Q families, and one polynomial-sign lemma in the odd-minimum region are computer-assisted results: the text states their exact inputs, domains, and acceptance conditions, while the source programs construct and check the finite polynomial obligations. Third, Proposition 2.2 is an exhaustive exact-integer computation over a declared finite set.

Every proof decision in the programs uses integer or rational arithmetic. The trusted high-level operations are polynomial factorization, gcd, resultants, and Sturm root counting in SymPy, together with GMP integer arithmetic. Hash files identify the reviewed source and output snapshot but do not themselves prove that a program was run. Section 7 gives the certificate rules, trust boundary, and theorem-by-theorem inventory.

2. Notation and recurrence

2.1. Notation

Throughout:

$$\begin{aligned}
&M, Q \in \mathbb{Z}_{\geq 0}; \quad D = M + Q; \quad x = M - Q \geq 0; \\
&p_s = [w^s]\,(1+w)^M (1-w)^Q \text{ (integers; } p_s = 0 \text{ off } [0, D]); \\
&s \text{ top-half index, } D/2 \leq s \leq D; \quad u = 2s - D \geq 0 \text{ (top-half offset)}; \\
&U = (u+1)^2; \quad E = (D+3)^2; \\
&\mathrm{Tur}(s) = p_s^2 - p_{s-1}p_{s+1}; \quad T_s = \mathrm{Tur}(s) - \mathrm{Tur}(s+1); \\
&k = D - s \text{ (distance from the top } = (D-u)/2).
\end{aligned}$$

Reflection: $p_{D-j} = (-1)^Q p_j$ for all j (substitute $w \to 1/w$). The top initial values are $p_D = (-1)^Q$ and $p_{D-1} = (-1)^Q x$. The sequence obeys the step-1 recurrence (contiguous in s; from the log-derivative of the generating function):

$$(s+1)\,p_{s+1} \;=\; x\,p_s + (s-1-D)\,p_{s-1}. \tag{2.1}$$

Lemma 2.1 *(no consecutive zeros)***.** *For $0 \leq j \leq D$, the two consecutive values p_j, p_{j+1} cannot both vanish.*

Proof. If $p_j = p_{j+1} = 0$, then (2.1) at index j gives $p_{j-1} = 0$ because $j - 1 - D \neq 0$ for $0 \leq j \leq D$. Iteration would give $p_0 = 0$, contrary to $p_0 = 1$. ∎

We write $\varepsilon = (-1)^Q$ and $\binom{n}{k}$ for binomial coefficients. In the gap sections (Sections 4–6), we relabel the exponents when necessary so that $0 \leq Q \leq M$; thus Q denotes the smaller exponent.

2.2. Finite verification

Proposition 2.2 *(finite verification)***.** *The inequality $T_s \geq 0$ holds for every $M, Q \geq 0$ with $D \leq 1200$ and every $\lceil D/2 \rceil \leq s \leq D$.*

Proof. The recurrence (2.1), initialized by $p_0 = 1$ and $p_1 = x$, was evaluated with GMP integers for all such triples. The computation examined 289,261,901 cells and found no negative value. Exact division was checked before every recurrence step. The recurrence-built values were checked against the defining binomial convolution for every coefficient with $D \le 12$ and at deterministic indices in six larger rows, reaching $D = 1200$. The program, output, and hash are listed in Section 7. ■

A separate Python computation verifies the recurrence for $M, Q \le 30$ and checks $T_s \ge 0$ directly for $M, Q \le 90$.

3. Quadratic form and regime decomposition

3.1. The bulk band

Theorem 3.1 *(quadratic-form identity and the bulk band)***.** *Eliminating p_{s+2} and p_{s-1} from T_s by the step-1 recurrence (2.1) gives the exact identity*

$$T_s = A\,p_s^2 + B\,p_s p_{s+1} + C\,p_{s+1}^2,$$
$$A = \frac{u+2}{s+2}, \qquad B = \frac{-x(u+1)}{(D+1-s)(s+2)}, \qquad C = \frac{u}{D+1-s}, \tag{3.1}$$

with discriminant

$$4AC - B^2 = \frac{u(u+2)\big((D+3)^2 - (u+1)^2\big) - x^2(u+1)^2}{\big((D+1-s)(s+2)\big)^2}. \tag{3.2}$$

Hence for $u \ge 1$, writing $U = (u+1)^2$, $E = (D+3)^2$ (note $u(u+2) = U-1$):

$$x^2 U \le (U-1)(E-U) \implies T_s \ge 0 \tag{3.3}$$

without further information about the sequence. The condition defines the bulk band $U \in [U_-, U_+]$, where

$$U_\pm = \frac{h \pm \sqrt{h^2 - 4E}}{2}, \qquad h := E + 1 - x^2,$$

are the roots of $U^2 - hU + E = 0$.

Proof. Substitute (2.1) for p_{s+2} and p_{s-1} in (1.2); simplification gives (3.1) and (3.2). The displayed roots are real throughout the parameter domain: since $0 \le x \le D$,

$$h = E + 1 - x^2 \ge (D+3)^2 + 1 - D^2 = 6D + 10 > 2(D+3) = 2\sqrt{E}.$$

If $u \ge 1$, then $A, C > 0$, and the quadratic form is positive semidefinite whenever $4AC - B^2 \ge 0$. ■

Remark (sharpness). The *sign* of the discriminant is basis-invariant: re-expressing T_s in a shifted basis multiplies it by a positive square. For example, the (p_{s-2}, p_{s-1}) discriminant is the displayed one times

$$\left(\frac{(D+1-s)(D+2-s)}{s(s+1)}\right)^2.$$

Thus no change among the shifted recurrence bases makes the form positive semidefinite outside the bulk band. The remaining arguments therefore use additional information from the Krawtchouk recurrence.

3.2. The center

Theorem 3.2 *(central identity, $u = 0$, D even)***.** *By reflection ($p_{D-j} = (-1)^Q p_j$): for Q odd, $p_{D/2} = 0$ and $T_{D/2} = 0$ identically. For Q even, the reflection constraint $p_{s-1} = p_{s+1}$ at $s = D/2$ collapses the recurrence to the rank-one relation $(D+2)\,p_{s+1} = x\,p_s$, and*

$$T_{D/2} \;=\; \frac{4\,p_{D/2}^2\left((D+2)^2 - x^2\right)}{(D+2)^2(D+4)} \;\geq\; 0, \tag{3·4}$$

with equality iff $p_{D/2} = 0$.

Proof. If Q is odd, reflection gives $p_{D/2} = 0$ and $p_{D/2-1} = -p_{D/2+1}$, which makes the four terms in $T_{D/2}$ cancel. If Q is even, reflection gives $p_{s-1} = p_{s+1}$ at $s = D/2$. Substitution in (2.1) yields $(D+2)p_{s+1} = xp_s$; inserting this relation into (1.2) and using the recurrence once more for p_{s+2} gives (3·4). ■

The rank-one central relation $(D+2)p_{c+1} = xp_c$ supplies the initial condition for the two recurrence arguments in Sections 5 and 6.

When M and Q are both odd, D is even and $T_{D/2} = 0$. This completely describes equality at the central step. If D is even, then M, Q have the same parity. The odd–odd case is the first case of Theorem 3.2; in the even–even case, $p_{D/2} = 0$ together with $(D+2)p_{D/2+1} = xp_{D/2}$ would give two consecutive zeros, contrary to Lemma 2.1. We do not claim a classification of equality at noncentral steps.

3.3. The tail: Riccati comparison flow

Theorem 3.3 *(tail flow)***.** *Let $k = D - s \geq 1$. Whenever its radicand is nonnegative, let*

$$w_s \;=\; \sqrt{x^2 - 4k(D+1-k)}$$

denote the nonnegative square root. On the flow region $w_s^2 \geq 0$, equivalently $(u+1)^2 \geq (D+1)^2 - x^2$ by the identity $(D+1)^2 - (u+1)^2 = 4(s+1)(D-s)$, one has $T_s \geq 0$.

Proof. The flow region with $k \geq 1$ is empty when $x = 0$, so assume $x > 0$. On the top half, $0 \leq k \leq D/2$, and $k \mapsto 4k(D+1-k)$ is increasing. Hence if the target index is in the flow region, so is every index between it and $D-1$; the following downward induction runs on this connected tail. The center $u = 0$ cannot lie in the flow region, because there $4k(D+1-k) = D(D+2) > D^2 \geq x^2$. Thus $u \geq 1$ and the quadratic-form coefficient C is positive throughout the tail.

Use the ratio variable $r_s = p_{s+1}/p_s$. The top initial values $p_D = (-1)^Q$, $p_{D-1} = (-1)^Q x$ give $r_{D-1} = 1/x$, and the recurrence gives the downward Möbius flow

$$r_{s-1} \;=\; \frac{D+1-s}{x - (s+1)\,r_s}.$$

Set $\phi_s = (x + w_s)/2$ and define the auxiliary downward Riccati flow $y_{s-1} = x - (s+1)(D-s)/y_s$, $y_{D-1} = x$. Then:

1. *(comparison)* The inequality $\phi_{s-1} \le x - (s+1)(D-s)/\phi_s$ reduces, after clearing denominators by $4(s+1)(D-s) = x^2 - w_s^2$, to $(w_s - w_{s-1})(x + w_s) \ge 0$. This holds because $w_s^2 - w_{s-1}^2 = 4u > 0$. Induction from $y_{D-1} = x \ge \phi_{D-1}$ gives $y_s \ge \phi_s > 0$ on the connected tail.

   Starting from $r_{D-1} = 1/x = (D-(D-1))/y_{D-1}$, a simultaneous induction in the two displayed recurrences gives $r_s = (D-s)/y_s = k/y_s$. Consequently $0 < r_s \le k/\phi_s$, every Möbius denominator is positive, and no p_s in the tail vanishes.

2. *(window value)* The quadratic-form value of Theorem 3.1 at $r = k/\phi_s$ is $I_1/\big((s+2)(D+1-s)\phi_s^2\big)$ with

$$\begin{aligned} I_1 &= (u+2)(k+1)\,\phi_s^2 - x(u+1)\,k\,\phi_s + u(D-k+2)\,k^2 = P_1 + Q_1 w_s, \\ Q_1 &= \tfrac{1}{2}\,x(D-k+2) \ge 0, \\ P_1 &= \tfrac{1}{2}\big[x^2(D+2-k) - 2k(D^2 - 2Dk + 3D + 2k^2 - 2k + 2)\big], \end{aligned} \tag{3.5}$$

   The coefficient $D+2-k$ of x^2 is positive. Thus P_1 is increasing in x^2, and at the flow boundary $x^2 = 4k(D+1-k)$ it equals $k(D+2)(u+1) > 0$. Since also $Q_1, w_s \ge 0$, this proves $I_1 \ge 0$ throughout the flow region.

3. *(window vertex)* Since $C > 0$, the inequality $k/\phi_s \le -B/(2C)$ reduces to

$$I_2 = x(u+1)\,\phi_s - 2u(D-k+2)\,k = P_2 + Q_2 w_s, \tag{3.6}$$

   where

$$Q_2 = \frac{x(u+1)}{2}, \qquad P_2 = \frac{x^2(u+1) - 4k(D^2 - 3Dk + 2D + 2k^2 - 4k)}{2}.$$

   Since $u+1 > 0$, P_2 is increasing in x^2; at the flow boundary it equals $2k(k+1) > 0$. Hence $I_2 \ge 0$.

4. Since $A, C \ge 0 \ge B$, the quadratic $A + Br + Cr^2$ is decreasing on $[0, -B/(2C)]$; by (1)–(3), $T_s/p_s^2 = A + Br_s + Cr_s^2 \ge$ (value at k/ϕ_s) ≥ 0. ∎

3.4. Coverage, and the gap

Corollary 3.4 *(coverage)*. *For $D = 0$ the only cell is $s = 0$ and $T_0 = 1$. For $D \ge 1$, every top-half cell is covered by the top, center, bulk, or flow arguments except the set*

$$\mathcal{G} = \big\{u \ge 1 : x^2 U > (U-1)(E-U), \quad U < (D+1)^2 - x^2\big\}, \tag{3.7}$$

equivalently $\{u \ge 1 : U < U_-\}$.

Proof. At the flow boundary $U_0 = (D+1)^2 - x^2$ the bulk margin is

$$(U_0 - 1)(4D+8) - x^2 = (D^2 + 2D - x^2)(4D+8) - x^2 > 0$$

for $0 \le x \le D$ (linear decreasing in x^2, with value $7D^2 + 16D$ at $x^2 = D^2$). Hence $U_- \le U_0 \le U_+$, and every top-half cell is covered by $\{s = D\}$ ($T_D = 1$), $\{u = 0\}$ (Theorem 3.2), the bulk ($U_- \le U \le U_0$), and the flow ($U \ge U_0$), except for (3.7). ∎

Lemma 3.5 *(heredity of the gap).* *If (D, x, u) is a gap cell and $u' \equiv u \pmod 2$ satisfies $1 \le u' \le u$, then (D, x, u') is also a gap cell.*

Proof. Corollary 3.4 identifies the gap exactly with $1 \le (u+1)^2 < U_-$. Since $(u'+1)^2 \le (u+1)^2$, the same inequality holds for u'. The parity condition makes $s' = (D+u')/2$ an integer index. ∎

Using $E - x^2 = (2Q+3)(2M+3)$, failure of the bulk inequality is equivalent to

$$(2Q+3)(2M+3) \quad < \quad U + \frac{E}{U} - 1. \tag{3.8}$$

Together with the non-flow condition in (3.7), this characterizes the gap. Thus gap cells occur only when the two exponents are unequal. An exact census shows that the first gap cell with $u \ge 4$ and $Q \ge 3$ appears at $D = 442$, $(Q, u) = (3, 4)$. Over all Q the first such cell is instead $(D, Q, u) = (142, 0, 4)$; it is covered by Proposition 4.3 before this residual region is considered.

The `regime-decomposition` program also checks this partition for every top-half cell with $D \le 90$.

Lemma 3.6 *(gap strip).* *After relabeling so that $Q \le M$, every gap cell satisfies*

$$(u+1)^2 = U < U_- \le \frac{7}{6}\,\frac{D+3}{2Q+3}. \tag{3.9}$$

Proof. Here

$$h = E + 1 - x^2 = (2Q+3)(2M+3) + 1,$$

and $h - 3(D+3) = 4MQ + 3M + 3Q + 1 > 0$. Thus $h > 3\sqrt{E}$. Since

$$U_- = \frac{2E}{h + \sqrt{h^2 - 4E}},$$

putting $t = h/\sqrt{E}$ shows that

$$\frac{U_-}{E/h} = \frac{2t}{t + \sqrt{t^2 - 4}}.$$

The last function is decreasing for $t \ge 2$, and its value at $t = 3$ is $6/(3+\sqrt{5}) \le 7/6$. Hence $U_- \le (7/6)E/h$. Finally $h \ge (2Q+3)(D+3)$ because $2M+3 \ge D+3$ when $Q \le M$. ∎

3.5. Coverage and thresholds

Table 1 records which statement covers each region. Theorems 3.1–4.2 and Propositions 4.1–4.3 cover every cell with $D \le 4586$; the first cell not covered by those results is $(D, Q, u) = (4587, 3, 15)$. Corollary 4.6 extends the threshold to $D \le 14826$, and Theorems 5.1 and 6.2 cover all remaining cells.

The numerical cutoffs are bookkeeping constants, not conjecturally sharp features of the inequality. The exact scan through $D = 1200$ absorbs the finite-offset nondegeneracy remainder $D < 1191$; the fixed-Q edge-flow onsets are smaller still (at most 93). In the final residual region, $x \le D - 26$ is simply $x = D - 2Q$ with $Q \ge 13$.

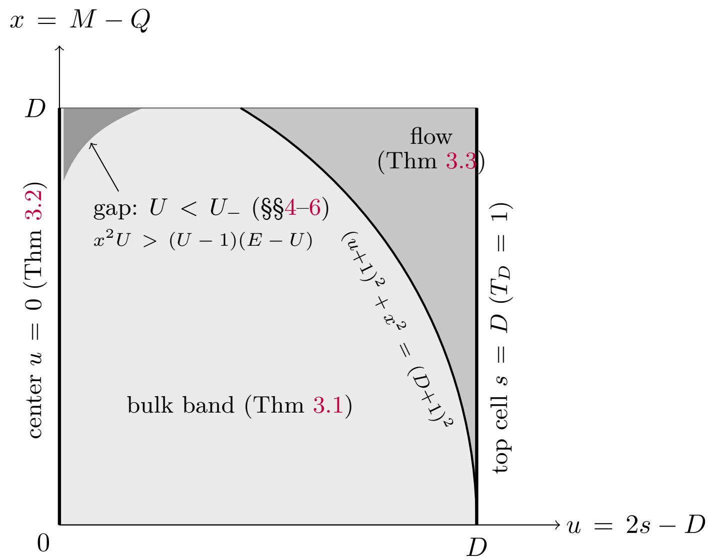

Figure 1. Coverage of the top half $\{2s \geq D\}$ in the (u, x)-plane for fixed D (schematic; lattice cells satisfy $u \equiv x \equiv D \pmod 2$). The bulk and flow regions are separated by the displayed circle. The dark gap is removed from the light bulk shading: it is the part of the non-flow region where the bulk quadratic form is indefinite, $\{u \geq 1 : U < U_-\}$. It is treated in Sections 4–6.

4. Reflection collapse and finite families

The finite part of the gap is treated in two ways: by fixed-offset families (Proposition 4.1 and Theorem 4.2) and by fixed-Q strips (Proposition 4.3 and Theorem 4.5).

4.1. Reflection collapse: gap offsets $u \leq 3$

Proposition 4.1 *(reflection collapse)***.** *Every gap cell with $u \leq 3$ has $T_s \geq 0$.*

Proof. For $s = (D+u)/2$ reflection gives $p_{s-u} = (-1)^Q p_s$; feeding this single constraint through the $(u+2)$-step recurrence chain determines the ratio p_{s-u+1}/p_{s-u} as an explicit rational function of (M, Q) and yields closed forms

$$T_s = R_u(M, Q)\ p_{s-u}^2.$$

Region or slice of the gap	Covered by
top cell $s = D$	trivial ($T_D = 1$)
center column $u = 0$	Theorem 3.2
bulk band $x^2U \le (U-1)(E-U)$	Theorem 3.1
flow region $U \ge (D+1)^2 - x^2$	Theorem 3.3
coverage: everything else = gap	Corollary 3.4
gap, $u \le 3$	Proposition 4.1
gap, $Q \ge 3$, $4 \le u \le 14$	Theorem 4.2
gap, $Q \le 2$ ($Q = \min$)	Proposition 4.3
gap, $3 \le Q \le 12$	Theorem 4.5
gap, $u \ge 15$, $Q \ge 13$, Q even	Theorem 5.1
gap, $u \ge 15$, $Q \ge 13$, Q odd	Theorem 6.2
all cells with $D \le 1200$	Proposition 2.2

Table 1. Coverage of the proof.

For $u = 1, 2, 3$, with $\varepsilon = (-1)^Q$ indicated by the superscript, the recurrence gives

$$R_1^+ = \frac{32\,(M+2)(Q+1)}{(D+3)^2(D+5)}, \qquad R_1^- = \frac{32\,(M+1)(Q+2)}{(D+3)^2(D+5)},$$

$$R_2^+ = \frac{16\,(M+1)(Q+1)\big[(M-Q)^2 + 2M^2 + 2Q^2 + 6M + 6Q\big]}{x^2(D+4)^2(D+6)},$$

$$R_2^- = \frac{32\,(M+2)(Q+2)}{(D+4)^2(D+6)},$$

$$R_3^+ = \frac{64\,(M+2)(Q+1)\big[(M-Q)^2 + 4Q^2 + 12Q - 1\big]}{(M-3Q-1)^2(D+5)^2(D+7)},$$

$$R_3^- = \frac{64\,(M+1)(Q+2)\big[(M-Q)^2 + 4M^2 + 12M - 1\big]}{(3M-Q+1)^2(D+5)^2(D+7)}$$

These expressions are nonnegative in the stated range: the two terms containing -1 are nonnegative for $Q \ge 1$ and $M \ge 1$, respectively, while for R_3^+ with $Q = 0$ one has $D = M \ge u = 3$ (a coefficient cell requires $u \le D$), so $M^2 - 1 > 0$. The only potentially negative residual values have $D < u$ and are not coefficient cells. The collapse denominators do not vanish on the gap: $x = 0$ fails the gap inequality outright; $M = 3Q + 1$ gives $16x^2 - 15(E - 16) = -(176Q^2 + 416Q - 16) < 0$ for $Q \ge 1$; and $3M - Q + 1 > 0$ for $M \ge Q$. Moreover, the seed p_{s-u} cannot vanish: once the displayed collapse denominator is nonzero, the reflection constraint and the recurrence force $p_{s-u+1} = 0$ as well, contradicting Lemma 2.1. Thus the ratio elimination used above is legitimate, and every gap cell with $u \le 3$ has $T_s \ge 0$. ■

4.2. Fraction-free constrained forms: gap offsets $4 \le u \le 14$

Theorem 4.2 *(finite gap offsets).* *Every gap cell with $Q \ge 3$ and $4 \le u \le 14$ has $T_s \ge 0$.*

Proof. For each u and parity class $\varepsilon = (-1)^Q$, the reflection constraint is encoded fraction-free: with the scaled chain $P_i = c_i\, p_{k_0+i}$ ($k_0 = s-u = D-s$; $c_0 = c_1 = 1$ and $c_{i+1} = c_i\,(k_0+i+1)$ for $i \ge 1$), the recurrence becomes

$$P_{i+1} = xP_i + (k_0+i-1-D)\frac{c_i}{c_{i-1}}P_{i-1}.$$

Thus every P_i has polynomial coefficients in the two seeds $(a,b) = (p_{k_0}, p_{k_0+1})$. All scale factors used below are positive: $k_0 \ge 0$ and every new factor k_0+i+1 is positive. In particular,

$$\widetilde{T} := T_s\, c_{u+1}c_{u+2} = \mathcal{A}a^2 + \mathcal{B}ab + \mathcal{C}b^2$$

is a polynomial quadratic form. The constraint $p_s = \varepsilon\, p_{k_0}$ reads $\lambda a + \mu b = 0$ with polynomial λ, μ, and pure algebra gives the two identities

$$\widetilde{T}\,\mu^2 = G\,a^2, \qquad \widetilde{T}\,\lambda^2 = G\,b^2, \qquad G := \mathcal{A}\mu^2 - \mathcal{B}\lambda\mu + \mathcal{C}\lambda^2, \tag{4.1}$$

at every cell satisfying the constraint, without division. Hence a cell is proved as soon as $G \ge 0$ there and $(\lambda,\mu) \ne (0,0)$. The `finite-gap-offsets` replay proves the following finite computer-assisted certificate for each (u,ε):

1. *(sector)* gap(u) with $Q \ge 3$ forces $Q < M/(u+1)^2$, by the estimate $2Q+3 < (D+3)/U + U/(D+3)$ with $(D+3)/U > 8$ (indeed $U < U_- \le D+3$ makes $\tau = (D+3)/U > 1$, and then $\tau + 1/\tau > 2Q+3 \ge 9$ forces $\tau > (9+\sqrt{77})/2 > 8$), and $M \ge 425$: the exact census already quoted in Section 3 (and repeated by this verifier) shows that no gap cell with $u \ge 4$ and $Q \ge 3$ has $D < 442$, so $Q < M/25$ gives $M > \frac{25}{26}D \ge \frac{25}{26}\cdot 442 = 425$ (hence $M \ge 426$; only $M \ge 425$ is used below).

2. *(two reduced constraints)* For the primary pinning let $g_1 = \gcd(\lambda_1,\mu_1)$ and strip $(\lambda_1,\mu_1) = g_1(\lambda_1',\mu_1')$, so $G_1 = g_1^2 G_1'$. The same construction for the alternate pinning $p_{s+1} = \varepsilon p_{k_0-1}$ gives (λ_2,μ_2), g_2, and G_2'. The two fraction-free identities (4.1) hold for each reduced constraint wherever its g_r is nonzero.

3. *(positivity)* $G' \ge 0$ on the sector via factorization: even-multiplicity factors are nonnegative; odd-multiplicity factors are positive because their leading homogeneous form $L(t)$ is positive on $[0, 1/(u+1)^2]$ (an exact 512-point partition and a derivative bound), while explicit weighted bounds control all lower-degree terms. This is done for both G_1' and G_2'. The resulting thresholds satisfy $M_0(u) \le 153$ through $u = 14$, so the sector bound $M \ge 425$ automatically lies beyond every positivity threshold.

4. *(nondegeneracy)* For *each* of the two pinnings, the reduced pair (λ_r', μ_r') has no common integer zero on the part of the sector with $M > 1145$ (resultant in Q, integer-root candidates, exact check). The omitted range is already covered by Proposition 2.2: since $Q < M/(u+1)^2 \le M/25$, $M \le 1145$ implies $D = M+Q < 1191 < 1200$. Where the first common factor g_1 vanishes, the alternative pinning $p_{s+1} = \varepsilon\, p_{k_0-1}$ takes over: a further resultant check proves $\{g_1 = 0\} \cap \{g_2 = 0\}$ contains no integer sector point with $M > 1145$, so there the second common factor is nonzero, the second reduced

identities are valid, and the second reduced pair is nonzero by its own nondegeneracy check above. (Nonvanishing of g_2 alone would not suffice: the conclusion also needs $(\lambda_2', \mu_2') \neq (0,0)$ at the cell.)

Here G' in item (3) denotes either reduced form G_r'. Consequently at least one reduced identity is valid at every cell, its right-hand side is nonnegative, and a nonzero reduced constraint coefficient forces $\widetilde{T} \geq 0$. Positivity of the scale factors then gives $T_s \geq 0$. The recursive inputs, generic sign lemmas, factor degrees, domination thresholds, and resultant-candidate reporting are stated explicitly in Section 7.2. ■

A worked finite-offset certificate. To make the computational mechanism inspectable without running the generator, consider $u = 4$ and $\varepsilon = +1$. After removing the common factor $g = M - Q = x$ from (λ, μ), the program obtains

$$\begin{aligned}
\lambda' &= -\frac{x(D+4)}{2}, \\
\mu' &= \frac{M^2 - 6MQ - 2M + Q^2 - 2Q}{2}, \\
G' &= \frac{(M+1)(Q+1)D^2(D+2)^2(D+4)^2}{128} F(M,Q),
\end{aligned}$$

where

$$\begin{aligned}
F(M,Q) &= 5M^4 - 20M^3Q + 78M^2Q^2 - 20MQ^3 + 5Q^4 \\
&\quad + 144M^2Q + 144MQ^2 - 20M^2 + 472MQ - 20Q^2.
\end{aligned}$$

On the certified sector put $t = Q/M < 1/25$. Dropping positive terms gives

$$\frac{F(M,Q)}{M^4} \geq \frac{13121}{3125} - \frac{20(1+t^2)}{M^2} > 0 \qquad (M \geq 425).$$

Moreover $x \neq 0$ on a gap cell, so $g \neq 0$ and $\lambda' \neq 0$; the alternate pinning and the resultant checks are therefore not needed for this family's argument (the verifier still performs them). The identities (4.1) therefore prove the $u = 4$, $\varepsilon = +1$ family directly. The other 21 offset–parity families follow the same construction, with the longer factor and nondegeneracy data recorded by the certificate generator.

Together with the preceding results and Proposition 4.3, this covers every offset $u \leq 14$ and every cell with $D \leq 4586$.

4.3. Per-Q closed forms: $\min(M,Q) \leq 2$

Proposition 4.3 *($\min(M,Q) \leq 2$, all D).* *If $\min(M,Q) \leq 2$, then $T_s \geq 0$ at every top-half index.*

Proof. Put $v := D - u = 2(D - s)$. For fixed Q, exact cancellation gives

$$\frac{T_s}{\binom{M}{s}^2} = \frac{N_Q(D,u)}{\Delta_Q(D,u)} \qquad (u \leq D - 10),$$

where $\Delta_Q > 0$: after factorization it is a positive constant times squares, positive factors $D+u+c$, and the positive factor $D+2-u$. (Equivalently, the unreoriented symbolic denominator has the single negative factor $u-D-2$.) The edge $v < 10$ is covered by the trivial top cell when $s = D$, and by Theorem 3.3 for $D \geq 24$ when $s < D$. The oriented numerators N_Q are:

$$\begin{aligned} N_0 &= (D+1)(D+2)(1+u), \\ N_1 &= D(D+1)\,u(u+1)(u+2), \\ N_2 &= D(D-1)\,S_2, \\ S_2 &= c_0 + c_1 u + c_2 u^2 + c_3 u^3 + c_4 u^4 + c_5 u^5 \\ &= 3D(D+2) + (D+2)(3D-4)\,u - 2(3D+4)\,u^2 \\ &\quad - 2(D-2)\,u^3 + 5u^4 + u^5. \end{aligned} \tag{4.2}$$

The $Q = 0$ numerator is positive; the $Q = 1$ numerator is nonnegative and vanishes exactly at $u = 0$ (Theorem 3.2, odd parity); and $S_2 \geq 0$ on the strip $u^2 \leq D/13$ by *paired domination*:

$$13c_0 + Dc_2 \;=\; D(33D+70) \;\geq\; 0, \qquad 13c_1 + Dc_3 \;=\; 37D^2 + 30D - 104 \;\geq\; 0,$$

with $c_4, c_5 \geq 0$ (the second paired polynomial is positive for $D \geq 2$, the only relevant $Q = 2$ range).

It remains to justify that this strip contains the whole $Q = 2$ gap. Here $x = D-4$, $E = (D+3)^2$, and $h = E + 1 - x^2 = 14D - 6$. Since $h \geq 2\sqrt{E}$ for $D \geq 3$, rationalizing the smaller root gives

$$U \;<\; U_- \;=\; \frac{2E}{h + \sqrt{h^2 - 4E}} \;\leq\; \frac{E}{h - \sqrt{E}} \;\leq\; \frac{(D+3)^2}{13D-9}.$$

Moreover

$$13(D+3)^2 \;\leq\; (D+13)(13D-9) \quad\iff\quad 82D - 234 \geq 0.$$

Consequently $(u+1)^2 \leq D/13 + 1$, and thus $u^2 = (u+1)^2 - 2u - 1 \leq D/13$ for $u \geq 0$. Off the strip Corollary 3.4 applies. The edge cells with $D < 24$ lie in Proposition 2.2. ■

4.4. Per-Q strip checks: $3 \leq Q \leq 12$

Lemma 4.4 *(fixed-argument certificate)***.** *Fix $3 \leq Q \leq 12$ and set*

$$R_j(M,s) \;:=\; \sum_{b=0}^{Q} (-1)^b \binom{Q}{b} \begin{cases} \displaystyle\prod_{i=0}^{j-b-1} \frac{M-s-i}{s+i+1}, & j-b \geq 0, \\ \displaystyle\prod_{i=0}^{b-j-1} \frac{s-i}{M-s+i+1}, & j-b < 0. \end{cases}$$

On the strip $0 \leq u \leq D - 2Q - 6$,

$$\frac{T_s}{\binom{M}{s}^2} \;=\; R_0^2 + R_0 R_2 - R_1^2 - R_{-1} R_1.$$

After substituting $M = D - Q$*,* $s = (D+u)/2$*, orienting the denominator, and removing positive content, write its numerator as* $S_Q = \sum_j c_j(D)u^j$*. On* $0 \leq u \leq D - 2Q - 6$ *the oriented denominator is positive. Every negative term* $c_j u^j$ *has the unique partner two degrees below, and for* $D \geq 2Q + 6$

$$B_Q(D) := \frac{7}{6}\frac{D+3}{2Q+3},$$
$$c_j(D) \leq 0, \qquad c_{j-2}(D) \geq 0, \qquad c_{j-2}(D) + c_j(D)B_Q(D) \geq 0.$$

All unpaired coefficients are nonnegative on the same ray.

Proof. The displayed finite sum follows by dividing the defining binomial convolution for p_{s+j} by $\binom{M}{s}$. Its denominator factors as a positive constant times squares and positive factors $D + u + c$, together with $D + 2 - u > 0$ on the stated interval. Exact expansion supplies the $c_j(D)$. Their signs and the partner inequalities are univariate polynomial claims on $[2Q + 6, \infty)$ and are certified by integer Sturm chains. For even Q, the partner indices $j - 2$ are $4r, 4r + 1$ $(0 \leq r < Q/2)$; for odd Q, they are $4r + 2, 4r + 3$ $(0 \leq r < (Q - 1)/2)$. These lists exhaust the negative coefficients. The exact generation and replay specification is given in Section 7.2. ∎

Theorem 4.5 *(per-Q strip checks)***.** *After relabeling so that* $Q = \min(M, Q)$*, the Main Theorem holds for every fixed-argument strip* $3 \leq Q \leq 12$*.*

Proof. On $0 \leq u \leq D - 2Q - 6$, Lemma 4.4 applies. Lemma 3.6 places every gap cell in $u^2 \leq B_Q(D)$. Hence each certified pair satisfies

$$c_{j-2}u^{j-2} + c_j u^j = u^{j-2}(c_{j-2} + c_j u^2) \geq u^{j-2}(c_{j-2} + c_j B_Q(D)) \geq 0,$$

and the unpaired terms are nonnegative. Thus $S_Q \geq 0$, and positivity of the oriented denominator gives $T_s \geq 0$ throughout this strip.

For the top edge $v < 2Q + 6$, the inequality defining the flow region holds once

$$(D - 2Q - 5)^2 \geq (2Q + 1)(2D - 2Q + 1).$$

For $Q = 3, \ldots, 12$, the exact respective onsets are

$$31,\ 38,\ 45,\ 52,\ 59,\ 66,\ 73,\ 80,\ 86,\ 93.$$

Thus the largest onset is 93; all earlier edge cells are covered by Proposition 2.2. Corollary 3.4 applies off the gap. Applying the `fixed-argument-strips` verification for $Q = 3, \ldots, 12$ proves the theorem. ∎

Corollary 4.6 *(finite-family coverage)***.** *The Main Theorem holds whenever* $\min(M, Q) \leq 12$*, whenever* $u \leq 14$*, and for every cell with* $D \leq 14826$*. The first gap cell left after the fixed-offset and fixed-argument families is*

$$(D, Q, u) = (14827, 13, 15)$$

after relabeling so that $Q \leq M$*.*

Proof. The assertion for $\min(M,Q) \le 12$ is Proposition 4.3 together with Theorem 4.5. The assertion for $u \le 14$ is Proposition 4.1, Theorem 4.2, and Proposition 4.3.

It remains only to identify the first cell outside both finite families. For each D, heredity of the gap means that only the first parity-compatible $u \ge 15$ need be tested. At fixed (D,u), the largest parity-compatible $x = D - 2Q$ with $Q \ge 13$ and

$$x^2 \; < \; (D+1)^2 - (u+1)^2$$

is the only candidate needed for the remaining strict bulk-failure inequality: if it fails, every smaller admissible x fails as well. Exact integer enumeration gives no candidate through $D = 14826$ and gives $(14827, 13, 15)$ at the next degree. The `fixed-argument-strips` generator records and repeats this census; the generic census rule is specified in Section 7. ■

5. Even-minimum gap cells

Theorem 5.1 *(even-minimum gap cells)***.** *After relabeling so that $Q \le M$, every gap cell with Q even and $u \ge 15$ has $T_s \ge 0$.*

Proof. Define (j runs over top-half indices)

$$d_j \;\; := \;\; (D+2)\, p_{j+1} - x\, p_j \quad \text{(deviation from the central rank-one relation)},$$

$$\Lambda \;\; := \;\; (D+2)^2 - x^2 \;\; = \;\; 4(Q+1)(M+1) \;\; > \;\; 0, \qquad \widehat{u}_j \;\; := \;\; 2(j+1) - D.$$

By Lemma 3.5, every positive parity-compatible offset between the center and the target remains in the gap; at the even-D center the same inequality below follows directly from $1 < (u+1)^2 < U_-$. Thus, for each index j used,

$$(2j-D)^2 \; < \; (2j - D + 1)^2 \; < \; (D+1)^2 - x^2,$$

so all pointwise oscillatory-zone estimates used below hold along the entire induction path. We use five steps.

1. *(Turán-step identity)*

$$T_j \;\; = \;\; \frac{\widehat{u}_j \ \mathrm{Tur}(j)}{j+2} \;\; - \;\; \frac{p_{j+1} d_j}{(j+2)(D+1-j)}. \tag{5.1}$$

   Since $\widehat{u}_j = 2(j+1) - D > 0$ at every top-half index used, $T_j \ge 0$ follows from $\mathrm{Tur}(j) \ge 0$ together with $p_{j+1} d_j \le 0$.

2. *(positivity of the Turán determinant)*

$$(D+1-j)\ \mathrm{Tur}(j) \;\; = \;\; (D+1-j)\, p_j^2 - x\, p_j p_{j+1} + (j+1)\, p_{j+1}^2$$

   is a quadratic form in (p_j, p_{j+1}) with discriminant $x^2 - \left[(D+2)^2 - (2j-D)^2\right]$ (using $4(j+1)(D+1-j) = (D+2)^2 - (2j-D)^2$), which is < 0 on the whole oscillatory zone $(2j-D)^2 < (D+1)^2 - x^2$. Therefore $\mathrm{Tur}(j) > 0$ there unless $(p_j, p_{j+1}) = (0,0)$, which is impossible by Lemma 2.1. The leading coefficient $D+1-j$ is positive because $j \le D$.

3. *((p, d) system)* First order, polynomial coefficients:

$$p_{j+1} \;=\; \frac{x\,p_j + d_j}{D+2}, \qquad d_{j+1} \;=\; \frac{D-j}{(j+2)(D+2)}\,\big(x\,d_j - \Lambda\,p_j\big).$$

4. *(comparison map)* $\zeta_j := -d_j/p_j$ obeys

$$\zeta_{j+1} \;=\; \kappa_j\,\frac{\Lambda + x\,\zeta_j}{x - \zeta_j}, \qquad \kappa_j \;=\; \frac{D-j}{j+2}\ (\le 1 \text{ above the center});$$

the map is increasing in ζ on $[0, x)$ and in κ, and the $\kappa = 1$ map is exactly tan-addition:

$$F\big(\sqrt{\Lambda}\,\tan a\big) \;=\; \sqrt{\Lambda}\,\tan(a + \theta^*), \qquad \tan\theta^* \;=\; \frac{\sqrt{\Lambda}}{x}.$$

Anchors from the reflection (Q even here, $p_{D-j} = p_j$): for D even, $d_c = 0$ at $c = D/2$ (the rank-one relation of Theorem 3.2), so $\zeta_c = 0$; here $p_c \neq 0$, since otherwise the rank-one relation would contradict Lemma 2.1. For D odd, $p_{c+1} = p_c \neq 0$ at $c = (D-1)/2$ gives, in one exact step,

$$\zeta_{c+1} \;=\; \kappa_c\,(D + 2 - x) \;=\; \kappa_c\,\sqrt{\Lambda}\,\tan(\theta^*/2)$$

where $\tan(\theta^*/2) = \sqrt{\Lambda}/(D + 2 + x)$. Comparison with the tan-addition majorant then gives

$$\zeta_{c+\ell} \;\le\; \sqrt{\Lambda}\,\tan\big((\ell - \delta/2)\,\theta^*\big), \qquad \delta \;=\; D \bmod 2,$$

as long as the argument stays below $\pi/2$. More precisely, the path is $j = c, c+1, \ldots, s$ when D is even and $j = c+1, c+2, \ldots, s$ when D is odd. On this path the induction invariant is

$$p_j \neq 0, \qquad 0 \le \zeta_j \le \sqrt{\Lambda}\tan\big((j - D/2)\theta^*\big) < x.$$

The displayed anchors establish its base case. If it holds at j, then $x - \zeta_j > 0$, so the (p, d) system makes p_{j+1} nonzero with the same sign as p_j; monotonicity of the comparison map gives the next upper bound, while its positive numerator gives $\zeta_{j+1} \ge 0$.

5. *(angular bound)* On the gap, $(u+1)^2\sin^2\theta^* < 7/3$ because

$$\begin{aligned}(u+1)^2\sin^2\theta^* &\le \frac{7}{6}\,\frac{D+3}{2Q+3}\,\frac{4(Q+1)(M+1)}{(D+2)^2}\\ &\le \frac{7}{3}\,\frac{(M+1)(D+3)}{(D+2)^2} \;<\; \frac{7}{3}.\end{aligned}$$

The first line uses Lemma 3.6; the second uses $(Q+1)/(2Q+3) \le \frac{1}{2}$, and the strict last inequality follows from $(D+2)^2 - (M+1)(D+3) = Q(D+3) + 1 > 0$. Since $x > 0$ in the gap, $\theta^* := \arctan(\sqrt{\Lambda}/x)$ lies in $(0, \pi/2)$. By Jordan's inequality ($\theta \le \frac{\pi}{2}\sin\theta$ on $[0, \pi/2]$) and $u \ge 15$ (which gives $(u+2)/2 \le \frac{9}{16}(u+1)$):

$$\frac{u+2}{2}\theta^* \;<\; \frac{9}{16}\,\frac{\pi}{2}\sqrt{\frac{7}{3}} \;<\; \frac{\pi}{2}.$$

Write $j = c + \ell$; then $2j - D = 2\ell - \delta$, and at the target $\ell - \delta/2 = u/2$. Therefore, at every step through $j = s$,

$$(\ell - \delta/2)\theta^* \le \frac{u}{2}\theta^* < \frac{\pi}{2} - \theta^*.$$

Since $x = \sqrt{\Lambda}\cot\theta^* = \sqrt{\Lambda}\tan(\pi/2 - \theta^*)$, the comparison bound is strictly below x. This closes the induction invariant at every index through the target; in particular, no denominator in the Riccati map vanishes or changes sign. Consequently p retains its sign and $p_{j+1}d_j \le 0$ from the first top-half cell through $j = s$. Steps (1) and (2) now give $T_s \ge 0$. ■

6. Odd-minimum gap cells and proof assembly

6.1. Odd-minimum gap cells

Lemma 6.1 *(odd-transition certificate)***.** *Let $D \ge 14827$, $Q \ge 13$, $u \ge 1$, $x = D - 2Q$, and suppose*

$$\frac{16(D+3)}{17} \le x \le D - 26, \qquad 24u^2 \le D + 3.$$

For each offset r whose gap window is real, put

$$g_r = \sqrt{x^2(r+1)^2 - r(r+2)\big(E - (r+1)^2\big)},$$
$$y_+(r) = \frac{(D+2)\big(x(r+1) + g_r\big)}{r(D+r+4)}, \qquad \kappa(r) = \frac{D-r}{D+r+4},$$
$$\Phi_r(y) = (1+\kappa(r))x - \kappa(r)\frac{(D+2)^2}{y}.$$

If the windows at u and $u+2$ are both real, then

$$y \ge y_+(u) \implies \Phi_u(y) \ge y_+(u+2), \qquad \Phi_u(\infty) = (1+\kappa(u))x \ge y_+(u+2).$$

In addition, when D is odd,

$$\frac{(D+2)(2x+D+1)}{D+3} \ge y_+(1).$$

Proof. This is the exact polynomial certificate used in the odd-minimum induction. For transparency, its symbolic obligations are specified here. Put

$$W_1 = x(u+1) + g_u, \quad K = \frac{x\big((u+1)D + u^2 + 9u + 12\big)}{D+u+4},$$
$$B_0 = u(D-u), \quad C_0 = (u+2)(D+u+6),$$
$$\mathrm{disc}_2 = x^2(u+3)^2 - (u+2)(u+4)\big(E - (u+3)^2\big),$$
$$X_{\min 2} = \frac{(u+2)(u+4)\big(E - (u+3)^2\big)}{(u+3)^2}, \qquad X_{\max} = (D-26)^2.$$

Writing $g_{u+2} = \sqrt{\mathrm{disc}_2}$, direct substitution gives

$$\begin{aligned}\big(\Phi_u(y_+(u)) - y_+(u+2)\big)\frac{W_1 u(D+u+4)(u+2)(D+u+6)}{D+2}& \\ = u(D+u+4)\big(KW_1 - B_0C_0 - g_{u+2}W_1\big).& \end{aligned} \tag{6.1}$$

Every cleared factor is positive. Moreover $D > u$, $y_+(u) > 0$, and

$$\Phi_u'(y) = \kappa(u)\frac{(D+2)^2}{y^2} > 0,$$

so it is enough to certify this boundary value.

After reducing g_u^2 to its defining polynomial, clear the denominator of K and define P_1, R_1 uniquely by

$$(D+u+4)^2\big[(KW_1 - B_0C_0)^2 - \mathrm{disc}_2\, W_1^2\big] = 4(u+2)\big(P_1 + xR_1g_u\big);$$

the left-hand side is a polynomial because $(D+u+4)K$ is, and the clearing factor is a positive square, so no sign conclusion below is affected. As polynomials in $X = x^2$, P_1 is concave, with leading coefficient $2(u+1)^2(u-D)(D+u+6) < 0$, and R_1 is decreasing, with coefficient $2(u+1)(u-D)(D+u+6) < 0$. The replay certifies

$$\begin{aligned} O_1 &= R_1(X_{\max}) \geq 0, \\ O_2 &= P_1(X_{\max}) \geq 0, \\ O_3 &= (u+3)^4 P_1(X_{\min 2}) \geq 0, \\ O_4 &= \frac{(D+u+4)^2}{u+2}(K^2 - \mathrm{disc}_2)\bigg|_{X = X_{\max}} \geq 0. \end{aligned}$$

The reality of the successor window and the bound on x place $X = x^2$ in $[X_{\min 2}, X_{\max}]$. For $u \geq 25$, the four obligations follow by coefficient inspection after $D = 24u^2 - 3 + \rho$, $u = z + 25$; for $1 \leq u \leq 24$, they follow after $D = 14827 + \rho$ (here $\rho, z \geq 0$ are slack variables). Thus concavity gives $P_1 \geq 0$ at both ends of the allowed X-interval and hence throughout it, while monotonicity gives $R_1 \geq 0$. Next, $Kx(u+1)$ is increasing in X, because its numerator $X(u+1)\big((u+1)D + u^2 + 9u + 12\big)$ has positive coefficients; the same numerator shows $K > 0$. At the gap boundary $X = u(u+2)\big(E - (u+1)^2\big)/(u+1)^2$ the value of $Kx(u+1) - B_0C_0$ is $4u(D+3)(u+2)^2/(u+1)$; equivalently, after multiplication by the positive factor $(u+1)(D+u+4)/(u(u+2))$, it is exactly $4(D+3)(u+2)(D+u+4)$. The strict gap inequality therefore gives $Kx(u+1) - B_0C_0 > 0$, so the squared inequality has the correct unsquared sign and proves the finite-state implication. Finally, after multiplication by the positive factor in O_4, $K^2 - \mathrm{disc}_2$ is linear decreasing in X, with coefficient $4(u-D)(D+u+6) < 0$. Thus O_4 and $K > 0$ prove the state at infinity.

For the odd-D base, clearing the radical makes the claimed inequality a concave quadratic in x. Its values at $x = 16(D+3)/17$ and $x = D - 26$ become coefficient-positive polynomials after $D = 14827 + \rho$. All reductions are identities in $\mathbb{Z}[D, u, X]$ and are replayed as described in Section 7.2. ∎

Theorem 6.2 *(odd-minimum gap cells)***.** *After relabeling so that $Q \le M$, every gap cell with Q odd (either D parity) has $T_s \ge 0$.*

Proof. By Theorem 4.2 and Corollary 4.6 we may assume $Q \ge 13$, $u \ge 15$, $D \ge 14827$. First record the bounds needed by the transition certificate. Lemma 3.6 gives

$$(u+1)^2 \le \frac{7}{6}\frac{D+3}{29}, \qquad 24u^2 \le \frac{168}{174}(D+3) < D+3.$$

Also $x = D - 2Q \le D - 26$. Since $U = (u+1)^2 \ge 256$ and $U \le (D+3)/24$, the gap inequality gives

$$\begin{aligned} x^2 &> \left(1 - \frac{1}{U}\right)(E - U) > \frac{8}{9}\left(E - \frac{D+3}{24}\right) \\ &= E\,\frac{24(D+3)-1}{27(D+3)} \ge E\,\frac{256}{289}, \end{aligned}$$

where the last inequality is equivalent to $24(D+3) \ge 289$. Hence

$$17x > 16(D+3).$$

For a top-half index j, write $r = 2j - D$. When $p_j \ne 0$, use

$$y_j := \frac{(D+2)\,p_{j+1}}{p_j},$$

and define

$$g_r^2 = x^2(r+1)^2 - r(r+2)\big(E - (r+1)^2\big), \qquad y_+(r) = \frac{(D+2)\big(x(r+1) + g_r\big)}{r(D+r+4)}.$$

By Lemma 3.5, every positive same-parity offset below the target has a real gap window, so g_r is real on the induction path. To include zeros without division, define the projective state

$$\mathcal{C}_r := \left\{(a,b) : a \ne 0,\ \frac{(D+2)b}{a} \ge y_+(r)\right\} \cup \{(0,b) : b \ne 0\}.$$

The second set is the single state $y = \infty$, and simultaneous zeros are excluded by Lemma 2.1. On finite states the recurrence is Möbius map

$$y_{j+1} = (1+\kappa_j)\,x - \kappa_j\,\frac{(D+2)^2}{y_j}, \qquad \kappa_j = \frac{D-j}{j+2},$$

increasing in y on $y > 0$.

In y-coordinates the quadratic form (3.1) of Theorem 3.1 reads, exactly, $G(y_s)\,p_s^2 = (s+2)(D+1-s)(D+2)^2\,T_s$. Thus, when $p_s \ne 0$, $T_s \ge 0$ iff y_s avoids the open window $(y_-(u), y_+(u))$ of the upward parabola

$$G(y) = u(s+2)\,y^2 - x(u+1)(D+2)\,y + (u+2)(D+1-s)(D+2)^2 \qquad (6.2)$$

Cells with $p_s = 0$ are immediate, since then $T_s = u\,p_{s+1}^2/(D+1-s) \geq 0$. On gap cells G has real roots $y_-(u) < y_+(u)$, and direct use of $s = (D+u)/2$ gives

$$y_+(u) = \frac{(D+2)\big(x(u+1)+g_u\big)}{u(D+u+4)}. \tag{6.3}$$

It is therefore enough to prove $(p_j, p_{j+1}) \in \mathcal{C}_{2j-D}$ from the first noncentral cell to s.

Reflection for odd Q is $p_{D-j} = -p_j$. For D even: $p_c = 0$ at $c = D/2$ and $p_{c+1} \neq 0$ by Lemma 2.1, and $y_{c+1} = 2(D+2)x/(D+4)$ exactly. For D odd: $p_c \neq 0$ because $p_{c+1} = -p_c$, and hence $y_c = -(D+2)$ at $c = (D-1)/2$, so $y_{c+1} = (D+2)(2x+D+1)/(D+3)$ exactly. The base inequality $y_{c+1} \geq y_+(\text{first window})$ holds: for D even (first window at $u = 2$), eliminating the square root reduces this to $x^2 \leq (D+4)^2$, which always holds; for D odd (first window at $u = 1$) it is the final assertion of Lemma 6.1.

At each subsequent offset, Lemma 6.1 maps both the finite and infinite parts of $\mathcal{C}_r$ into $\mathcal{C}_{r+2}$. Induction therefore gives $(p_j, p_{j+1}) \in \mathcal{C}_{2j-D}$ along the whole strip. At the target, either $p_s = 0$ and the displayed formula for T_s is immediate, or $y_s \geq y_+(u)$ and hence $G(y_s) \geq 0$. Thus $T_s \geq 0$. Cells with $D \leq 14826$, or $Q \leq 12$, or $u \leq 14$ were already covered by Corollary 4.6. ■

6.2. Assembly: proof of the Main Theorem

Proof of Theorem 1.1 and Corollary 1.2. Fix $M, Q \geq 0$, put $D = M + Q$, and choose $D/2 \leq s \leq D$. By swapping the exponents if necessary, assume $x = M - Q \geq 0$ (Section 1.5). Write $u = 2s - D$ and $U = (u+1)^2$.

1. If $s = D$: $T_D = p_D^2 = 1 > 0$.
2. If $u = 0$: Theorem 3.2.
3. If $x^2U \leq (U-1)(E-U)$: Theorem 3.1 (bulk).
4. If $U \geq (D+1)^2 - x^2$: Theorem 3.3 (flow).
5. Otherwise the cell is a gap cell (Corollary 3.4). After relabeling, $Q = \min(M, Q)$:
   - $u \leq 3$: Proposition 4.1;
   - $Q \leq 2$: Proposition 4.3;
   - $4 \leq u \leq 14$: Theorem 4.2;
   - $3 \leq Q \leq 12$: Theorem 4.5;

   If none of these cases applies, then $u \geq 15$ and $Q \geq 13$. Corollary 4.6 also gives $D > 14826$. Theorem 5.1 applies when Q is even, and Theorem 6.2 applies when Q is odd.

Every branch ends in $T_s \geq 0$. The corollaries follow by telescoping from $\mathrm{Tur}(D) = 1$ and by the reflection symmetry of Tur (Section 1.1). ■

7. Machine verification

7.1. Methodology

The proof is computer-assisted in the following precise sense. The symbolic identities and finite positivity claims used in Sections 2–6 have exact-arithmetic replays: rational arithmetic (SymPy) for symbolic identities, integer Sturm chains for univariate positivity claims, and GMP integers for the exhaustive scan. No floating-point comparison or unquantified asymptotic estimate is used to prove the Main Theorem; the proof programs are float-free. In particular the orbit tan-bound comparison in the even-minimum verifier is exact: every orbit angle is a nonnegative multiple of θ^* (plus $\theta^*/2$ for odd D), where the tangent is a rational multiple of $\sqrt{\Lambda}$, and tan-addition preserves that form. The verification patterns are:

- *coefficient-positive polynomials*: after explicit integer substitutions (e.g. $D = 24u^2 - 3 + \rho$, $u = z + 25$ in Lemma 6.1) the certified polynomials have all coefficients nonnegative, so positivity is inspection;
- *Sturm counts* on explicitly given intervals for the univariate cases (per-Q pairing thresholds, interval base case, finite strips);
- *threshold censuses*: because the gap is $U < U_-$, only the first parity-compatible offset above a declared cutoff need be tested. At fixed (D, u) the admissible values $x = D - 2Q$ have the parity of D, and the two gap inequalities are exactly
$$\frac{(U-1)(E-U)}{U} < x^2 < (D+1)^2 - U.$$
Thus the largest parity-compatible x below the upper bound and the prescribed Q cutoff decides existence exactly;
- *exhaustive exact scans* for the remaining finite regions ($M, Q \leq 90$; $D \leq 1200$, 289,261,901 cells), through the certified recurrence, with divisibility checked before every GMP exact division. The resulting values are cross-checked against every binomial coefficient for $D \leq 12$ and against deterministic coefficients in six larger rows up to $D = 1200$. The cell count is asserted in-program against
$$\sum_{D=0}^{1200} (D+1)\big(\lfloor D/2 \rfloor + 1\big) = 289{,}261{,}901;$$
- *end-to-end consistency checks* confirming that the declared regions cover all cells with $D \leq 90$ and that each argument's output matches direct evaluation at explicit cells.

The committed text files are deterministic replay logs, not separate proof objects: the proof obligations and their exact checks are specified in the corresponding source programs. In particular, elapsed wall-clock times are excluded from the logs. The manifest hashes bind each log to the source program declared to have produced it; a hash establishes snapshot integrity, not that the program was executed. Acceptance of a computer-assisted step requires inspection or replay of that program, not merely inspection of a `PASS` line. The finite-offset log additionally records, for every offset and parity, the factor-degree pattern, domination threshold, resultant degree and integer-candidate count, and the number of exact spot cells tested in each reflection parity. The fixed-argument log records every paired coefficient index and its certified onset. The odd-minimum verifier checks the fully cleared transition identity (6.1) before any endpoint positivity test.

Thus the human-readable text is logically complete only together with the certificate-generating sources. The trusted base consists of the language runtimes and exact-arithmetic libraries listed in Section 7.3. In particular, the proof trusts SymPy's implementations of polynomial factorization, gcd, resultant, and Sturm root counting. The elementary rules below explain why the outputs requested from those implementations imply the required signs and nondegeneracy statements; the present logs do not constitute low-level certificates for the implementations themselves. The file `certificates/CERTIFICATE_MAP.md` maps every machine-dependent theorem to its generator, exact log, proof obligations, and replay command. The companion `certificates/REVIEW_CHECKLIST.md` gives a function-by-function inspection route: it identifies the constructor, sign test, nondegeneracy test, threshold census, and final acceptance predicate used by each verifier. This separates review of the mathematical implication from the diagnostic examples printed by the programs.

7.2. Certificate specifications

For clarity, we state here the elementary certificate rules used by the programs. They reduce the machine-assisted portions to finite lists of integer/rational arithmetic assertions.

Lemma 7.1 *(sign-certificate rules)***.** *The following are valid exact certificates.*

1. *Let $f(M,Q)$ have total degree d, put $t = Q/M$, and write*
$$M^{-d}f(M,Mt) \;=\; L(t) + \sum_{\nu\geq 1} M^{-\nu}L_\nu(t).$$
On $0 \leq t \leq t_0$, suppose a mesh of width t_0/N has minimum m_L for L, and $|L'| \leq B$. Then $L(t) \geq m_L - Bt_0/(2N) =: \ell$. If $|L_\nu(t)| \leq S_\nu$ and $\sum_{\nu\geq 1} S_\nu M^{-\nu} < \ell$, then $f(M,Q) > 0$.
2. *If $F \in \mathbb{Z}[D]$ has positive leading coefficient, has no real root in $[D_0,\infty)$ by its exact Sturm chain, and $F(D_0) > 0$, then $F(D) > 0$ for all $D \geq D_0$.*
3. *After clearing nonzero rational content, if $f,g \in \mathbb{Z}[M,Q]$ have a common integer zero, then its M-coordinate is an integer root of $\mathrm{Res}_Q(f,g)$. Factoring this univariate resultant over $\mathbb{Z}$, checking every linear-factor root, and then computing the specialized gcd in $\mathbb{Z}[Q]$ is therefore exhaustive.*

Proof. Part (1) is the mean-value theorem on the nearest mesh point followed by the triangle inequality. Part (2) is the intermediate-value theorem together with the Sturm root count. Part (3) is the defining vanishing property of the resultant; every integer root of a univariate integer polynomial occurs as a rational linear factor. ∎

Finite offsets. The inputs are not opaque expanded expressions: for each $u \in \{4, \dots, 14\}$ and $\varepsilon \in \{\pm 1\}$ they are defined recursively by the scaled chain in Theorem 4.2, followed by

$$(\lambda, \mu) = \mathrm{coeff}_{a,b}(P_u - \varepsilon c_u a), \qquad G = \mathcal{A}\mu^2 - \mathcal{B}\lambda\mu + \mathcal{C}\lambda^2.$$

The program constructs these objects in $\mathbb{Q}[M, Q]$, verifies both identities in (4.1), removes $g = \gcd(\lambda, \mu)$, and applies Lemma 7.1(1) with $N = 512$ and $t_0 = (u+1)^{-2}$. The core factor degrees and largest domination onsets are:

u	4	5	6	7	8	9	10	11	12	13	14
deg(+)	4	4	6	6	8	8	10	10	12	12	14
deg(−)	2	4	4	6	6	8	8	10	10	12	12
M_0	4	9	13	25	33	50	63	85	103	130	153

Here deg(±) refers to $\varepsilon = \pm 1$; the alternate pinning has the same core degree. The committed log gives every factor multiplicity, resultant degree, and integer-candidate count, while the source enumerates and checks the complete candidate set and specialized gcds prescribed by Lemma 7.1(3). The verifier also repeats, as an exact heredity-reduced integer scan, the census fact that no gap cell with $u \geq 4$ and $Q \geq 3$ has $D < 442$ (the input to the sector bound $M \geq 425$), and evaluates each constrained identity at the first gap cells of both reflection parities, reporting the tested cell counts.

Fixed arguments. The finite binomial-ratio input and denominator orientation are stated in Lemma 4.4. After expansion, for even Q the partner indices $j-2$ are $4r, 4r+1$ for $0 \leq r < Q/2$; for odd Q, they are $4r+2, 4r+3$ for $0 \leq r < (Q-1)/2$. Lemma 7.1(2) certifies all coefficient signs and $c_{j-2} + c_j B_Q(D) > 0$ from the common onset $D_0(Q) = 2Q + 6$; every unpaired coefficient is certified positive on the same ray. This is the complete pairing list for $3 \leq Q \leq 12$.

Odd-minimum endpoints. The four polynomial obligations in Lemma 6.1 are, in the notation there,

$$O_1 = R_1(X_{\max}), \qquad O_2 = P_1(X_{\max}),$$
$$O_3 = (u+3)^4 P_1(X_{\min 2}), \quad O_4 = (D+u+4)^2 \left.\frac{K^2 - \mathrm{disc}_2}{u+2}\right|_{X = X_{\max}}.$$

After $D = 24u^2 - 3 + \rho$, $u = z + 25$, their respective term counts and least coefficients are

	O_1	O_2	O_3	O_4
# terms	24	48	77	30
min coefficient	2	58	1	1

and hence all four are nonnegative for $u \geq 25$. For $1 \leq u \leq 24$, substituting $D = 14827 + \rho$ gives nonnegative coefficients directly; no Sturm fallback is used. These counts and minima are recomputed, not hard-coded, by the odd-minimum verifier.

7.3. Availability and trust boundary

Development source is maintained at https://github.com/skypher/kraw. The exact reviewed source is the submission artifact carrying `certificates/MANIFEST.sha256`, which binds the manuscript, programs, and deterministic outputs byte-for-byte; no claim is made that an untagged development branch reproduces that snapshot. The tagged submission snapshot is the immutable release `v1.0.2` at https://github.com/skypher/kraw/releases/tag/v1.0.2; its assets include the source archive, manuscript PDF, hash records, and path-sanitized profiled replay transcript. Before publication, the exact accepted snapshot must be deposited as an immutable tagged release and its archival DOI added to the final bibliographic record. Because the source programs are part of the proof, this deposit is an acceptance condition, not an optional post-publication convenience; an accepted version should not refer only to a moving development branch.

To avoid a circular identifier (the full manifest includes the manuscript), `certificates/PAYLOAD.sha256` separately hashes only the proof generators, exact logs, dependency pin, certificate map, and referee checklist. Its own SHA-256 digest is

`240ac277f430249bb08982d643c2c779d9cc08095ad21316436232fde45ce8ac`

This digest identifies the proof payload printed with this submission; the full manifest additionally binds the article and the PDF built in the reference toolchain. The repository contains `CITATION.cff` and release metadata ready for an archival deposit. The DOI cannot be created by the proof and must be added to the final version after the external deposit.

The trusted base is Python 3, SymPy 1.12, a C++17 compiler, and GMP; the programs themselves remain subject to inspection. Specifically, exact arithmetic does not remove reliance on SymPy's factorization, gcd, resultant, and Sturm algorithms. The proof programs contain no floating-point computation: every acceptance predicate, including the end-to-end orbit checks, is integer or rational arithmetic.

The command `make toolchain-info` prints the Python, SymPy, compiler, GMP, pdfTeX, and platform versions of a replay environment. `make audit-fast` combines the manifest checks, all five short exact replays, and a compiled GMP scan through $D = 120$. These commands do not replace the complete replay, but they expose environment drift and exercise every short proof mechanism before the multi-hour jobs are attempted. The command `make replay-profile` runs the complete replay under `/usr/bin/time -v`; the archival release should include the resulting wall time and peak resident memory together with the toolchain report. Resource values are machine-dependent and are therefore not embedded in the deterministic proof logs.

The Make targets `payload-check` and `verify` check stored hashes only; they do not execute a proof program. The target `replay-short` runs the five short exact verifiers, while `replay-all` additionally runs the two multi-hour symbolic jobs and the exhaustive GMP scan. Reference software versions and the individual commands are recorded in `paper/README.md`.

Artifact	Verifies
`recurrence-and-small-scan`	Recurrence identity ($M, Q \le 30$) and $T_s \ge 0$ exhaustively for $M, Q \le 90$.
`exhaustive-scan-D1200`	Exhaustive exact scan of 289,261,901 cells with zero negatives for all $D \le 1200$, including per-step divisibility tests and deterministic convolution audits (Proposition 2.2).
`regime-decomposition`	Theorems 3.1–3.3, Corollary 3.4, Proposition 4.1 (symbolic identities, grids, end-to-end coverage $D \le 90$, gap census).
`finite-gap-offsets`	Theorem 4.2: offsets $u = 4, \dots, 14$, both parities; sector estimate; threshold $D = 4586$.
`small-argument-cases`	Proposition 4.3: $Q \le 2$ for all D.
`fixed-argument-strips`	Theorem 4.5 and Corollary 4.6: per-Q strips $Q = 3, \dots, 12$; threshold census $D = 14826$.
`even-minimum-gap`	Theorem 5.1: even-minimum gap cells, all D (symbolic identities, angular bound, end-to-end checks for both D parities).
`odd-minimum-gap`	Theorem 6.2: odd-minimum gap cells (symbolic interval algebra, endpoint and finite-strip coefficient positivity, end-to-end recurrence checks near $D = 14828$).

Table 2. Verification inventory.

7.4. Verification inventory

Each row names both a program in `scripts/` (with suffix `.py`) and its committed output in `certificates/` (with suffix `.txt`); the exhaustive scan program is in `cpp/`. The accompanying `certificates/MANIFEST.sha256` records the exact artifact hashes, and `certificates/README.md` states the artifact trust boundary and replay convention.

8. Related work

The classical Turán inequality (Szegő [Sze48], for Legendre polynomials, answering P. Turán) asserts $P_n(x)^2 - P_{n-1}(x)P_{n+1}(x) \ge 0$ on the orthogonality interval; its descendants form a large literature on the *positivity* of Turán determinants as functions of the argument x, with the degree window fixed, for broad classes of orthogonal polynomials (for general background see [Sze75]). Five strands are closest to the present theorem.

Recurrence-based positivity criteria. Szwarc [Szw98] gave general criteria (monotonicity conditions on the three-term recurrence coefficients) under which orthogonal polynomials satisfy the Turán inequality; Berg and Szwarc [BS09] proved two-sided bounds on normalized Turán determinants, and Szwarc [Szw21] extended the positivity criteria further. Krasikov [Kra11] developed Turán inequalities adapted to three-term recurrences with monotone coefficients. Recent general criteria and nonnegative representations are developed in [Kah26]; that work also propagates positivity between determinant indices under recurrence hypotheses. These results control the *sign* or propagated sign of Tur, not the ordered comparison $\mathrm{Tur}(s) \ge \mathrm{Tur}(s+1)$ asserted here. The present integer arguments also belong to the Krawtchouk orthogonality support, so the location of the argument is not the distinction. Moreover, the cited general criteria are normally formulated after a prescribed orthogonal normalization. As explained in Section 1.4, degree-dependent rescaling changes the present monotonicity question; no deduction between the normalized and coefficient-normalized statements is being asserted.

Krawtchouk bounds. Krasikov (partly with Litsyn) developed Turán-type and Laguerre-type inequalities for discrete orthogonal polynomials and used them for sharp, parameter-uniform bounds on binary Krawtchouk polynomials, their extreme zeros, and Christoffel–Darboux kernels [Kra03, Kra05, KL96, Kra01]. Most directly, [Kra05, Theorem 5 and (17)] includes binary Krawtchouk polynomials in a Laguerre–Pólya generating-function framework and gives a fourth-degree refinement coupling two adjacent Turán determinants. That framework is written for exponential coefficients $\sum_k u_k z^k/k!$; converting it to the ordinary generating coefficients in (1.1) introduces a degree-dependent factorial normalization. Moreover, its conclusion controls positivity and a product of adjacent determinants, not the ordered difference $\mathrm{Tur}(s) - \mathrm{Tur}(s+1)$ in the present coefficient normalization. Thus it does not imply Theorem 1.1, but it is the closest explicit antecedent within the Krawtchouk literature and should not be conflated merely with an envelope estimate. The proof here instead uses the exact arguments of Sections 5–6.

Specialized Schur functions and character values. Proposition 1.4 identifies $\mathrm{Tur}(s)$ with the rectangular character value at the order-two element $\mathrm{diag}(I_M, -I_Q)$. Ayyer and Kumari [AK22, Theorem 2.5] factor general-linear characters under balanced root-of-unity specializations. Karmakar [Kar25, Theorem 4.1] gives formulas at arbitrary order-two conjugacy classes, including the present involutions; outside the factorized cases these formulas are alternating sums of products of dimensions. They compute individual character values but do not compare the rectangles (2^s) and (2^{s+1}), nor do they imply their positivity in the present specialization. Reiner and Stanton [RS98] prove symmetry and unimodality for certain differences of *principally* specialized Schur functions $s_\lambda(1, q, \dots, q^n)$. Their specialization and varying shapes differ from the fixed $\{1, -1\}$ alphabet and rectangle-height sequence here, so neither result implies the other. These works nevertheless place the Main Theorem alongside character specialization and Schur unimodality, rather than only classical Turán inequalities.

Coefficient and asymptotic inequalities. Turán inequalities for coefficient sequences and Jensen polynomials have a separate classical theory, for example Craven–Csordas [CC89]. Uniform asymptotics for Krawtchouk polynomials in the oscillatory regime are also available [LW00]. Neither line of work treats monotonicity of the degree-index Turán determinant at a fixed Krawtchouk argument.

Coding-theoretic Krawtchouk analysis. Binary Krawtchouk polynomials at integer arguments are the spectral kernel of the Hamming scheme [Del73, Lev95, MS77]; positivity of Krawtchouk quadratic expressions at integer points is central to linear-programming bounds. We are not aware of the specific quantity $\mathrm{Tur}(s) = p_s^2 - p_{s-1}p_{s+1}$ (in the coefficient/degree index, at a fixed integer argument) having been studied there.

We did not find the monotonicity statement of Theorem 1.1 in the orthogonal-polynomial, coding-theoretic, or specialized-character literature. Our novelty claim is limited to that precise statement for the coefficient normalization of the two-parameter family $(1 + w)^M(1 - w)^Q$ at integer (M, Q), equivalently the ordered rectangular character values in (1.3).

9. Further questions

1. **Uniform-Q pairing structure.** The per-Q strip certificates (Theorem 4.5) exhibit a recurring pattern: after removing positive content, the negative coefficients of S_Q are paired with positive neighbors two degrees below. A contiguity recursion in Q ($p_s^{(Q+1)} = p_s^{(Q)} - p_{s-1}^{(Q)}$) that provably preserves the paired cone would treat the entire gap uniformly and give a second, asymptotics-free proof of the Main Theorem.

2. **Beyond the symmetric weight.** Does Turán-step monotonicity persist for $[w^s]\,(1+aw)^M(1-bw)^Q$ (general (p,q)-Krawtchouk values, $a \neq b$), or for other products of two real-rooted polynomials whose roots interlace across 0? The reflection symmetry used for the central initial values and the finite recurrence identities is special to $a = b$; which parts of the argument survive its loss?

3. **Combinatorial meaning of** $\mathrm{Tur}(s) \geq 1$**.** On the top half the Turán determinant is not just nonnegative but ≥ 1 (integer telescoping). Is there a lattice-path or coding-theoretic injection realizing $p_s^2 - p_{s-1}p_{s+1} - 1$ as a counting quantity at integer arguments?

4. **Equality away from the center.** Theorem 3.2 classifies central equality, while the proof of the Main Theorem does not track equality through every certificate branch. Are there any noncentral cells with $T_s = 0$?

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